Vinujen Dilogen

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Technical Skills

Languages: Python, C/C++, SQL, JavaScript
AI/Machine Learning and Frameworks: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, LangChain, FastAPI, Flask, React
Developer Tools: Git, Docker, AWS, PostgreSQL, VS Code, Jupyter Notebook, pytest

Education

Bachelor of Engineering in Software Engineering Expected Graduation, May 2027
Ontario Tech University *Oshawa, ON*

- **President's List:** Fall 2025
- **Relevant Coursework:** Deep Learning Architecture, AI Engineering & MLOps, Distributed Systems, Cloud Computing, Data Management

Experience

Data and Process Analyst October 2026 – Present
Ontario Tech University – Office of Student Life *Oshawa, ON*

- Develop data analytics and visualization solutions using Python and Microsoft Power BI to analyze datasets, build interactive dashboards, and support data-driven decision-making
- Explore AI-powered tools and workflow automation to streamline repetitive tasks, optimize institutional processes, and improve operational efficiency
- Apply Python programming, data manipulation, and analytical techniques to develop technical solutions supporting institutional projects
- Collaborate with cross-functional teams and stakeholders to identify reporting requirements, improve data accuracy, and deliver actionable insights

Customer Service Manager May 2024 – August 2026
Walmart Canada *Stouffville, ON*

- Coordinate daily workflows for a team of customer service associates, prioritizing tasks, communicating operational requirements, and maintaining consistent execution in a high-volume environment

Projects

Vision-Language-Action (VLA) Excavation System September 2026 – Present

- **Capstone Project** under the supervision of Dr. Jaho Seo.
- Designing an AI-powered robotic excavation system integrating computer vision, large language models, and motion planning for autonomous task execution.
- Developing modular backend software in Python using PyTorch, OpenCV, ROS2, and LangChain while collaborating with a multidisciplinary engineering team.

AI Stock Market Predictor & Analyst | *Python, FastAPI, React, PostgreSQL* Jan. 2026 – Apr. 2026

- Architected a full-stack stock market prediction platform and led development of the machine learning pipeline.
- Implemented Linear Regression from scratch using Gradient Descent, feature engineering, and five years of historical market data.
- Developed a FastAPI backend and engineered financial indicators, reducing prediction error by approximately 10% compared to the baseline model.

Battery State of Health (SOH) Prediction Platform | *Python, Flask, React* Sept. 2025 – Dec. 2025

- Led a team of four students in developing a full-stack AI-powered lithium-ion battery health prediction platform.
- Designed a Linear Regression pipeline to estimate battery State of Health and deployed a Flask API with a React frontend on AWS.
- Integrated the OpenAI API for AI-assisted diagnostics and evaluated model performance using R^2, MAE, MSE, and RMSE.